

Pressure optimization control of low-pressure vacuum systems based on anti-disturbance Smith predictor compensation

Lulu Wu¹, Licheng Huang¹, Yuan Wang² , Benchu Jiang¹ and Dazhu Li¹

Abstract

In addressing the challenges posed by significant time delays and multiple disturbances in low-pressure vacuum systems, this paper proposes an anti-disturbance Smith predictor compensation control method. The method, based on the dynamic characteristics of the low-pressure vacuum system, integrates the synergistic advantages of feedforward and feedback control to design a composite controller with parameter self-adaptive tuning (AFFPI), which optimizes control parameters according to changes in the process state. Secondly, to address the challenge posed by the inherent lag characteristics of the vacuum system on control performance, a Smith predictor (SP) structure based on dynamic compensation was established in the internal loop, effectively solving the issue of slow response caused by significant time delays in the low-pressure vacuum system. Finally, an anti-disturbance filtering unit was introduced into the pressure feedback channel. By configuring appropriate filter gain factors, the unit alleviates fluctuations in the control signal caused by disturbances to the SP, preventing the performance degradation typically observed in traditional SP control methods under disturbances, thus achieving efficient and high-precision pressure regulation in the low-pressure vacuum system. The simulation results demonstrate that the proposed control method reduces the average settling time by 31.73 s, reduces the overshoot by 10.59%, and lowers the ITAE index by 1619.41 when compared to AFFPI, Mac-PID, and PID-Smith. Under four types of simulated disturbances, the method demonstrates the most optimal control performance, exhibiting an average enhancement of 19.66% in disturbance control performance in comparison to the Smith predictive control method that does not consider disturbances (AFFPI-Smith). The experimental findings demonstrate that the proposed method reduces the average settling time by 7.32 s and decreases the CV, TV, ITE, and ITAE indices by 1.8247×10^{-5} , 0.385, 3.85, and 26.23, respectively. This results in improvements in control accuracy, control time, and stability. Specifically, compared with AFFPI-Smith, the ITAE is reduced by 4.72, and the disturbance performance is improved by 11.12%. The proposed method maintains excellent control performance even when process conditions or control target pressures are adjusted, and it demonstrates strong robustness and disturbance rejection capability.

Keywords

low-pressure vacuum system, anti-disturbance control, significant time delay, Smith predictor, perturbation filter unit

Date received: 26 April 2025; accepted: 30 July 2025

Introduction

In the contemporary industrial manufacturing environment, low-pressure vacuum systems have emerged as a pivotal technology for establishing precise process conditions, a consequence of their high efficiency, stability, and reliability. Beyond generating a pristine and manageable low-pressure environment, these systems have been shown to markedly enhance production efficiency, material purity, material performance, and product consistency. As a result, these systems play a critical role in traditional areas such as vacuum coating,^{1–4}

precision casting,^{5–7} and high-performance composite manufacturing.^{8,9} With continuous technological

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breakthroughs, their application scope has rapidly expanded into emerging fields such as new energy materials development^{10,11} and aerospace materials preparation,^{12,13} providing strong support for the technological upgrading and sustainable development of modern industry.

In low-pressure systems, the precise and rapid regulation of pressure is crucial to the stability and efficient operation of the system. Especially in various industrial applications, the primary task of low-pressure control systems is to maintain pressure balance within equipment or processes, ensuring that the system operates within an ideal low-pressure environment. This stable low-pressure environment is not only the foundation for ensuring production efficiency but also directly impacts the safety of the production process and the quality of the final product. Any fluctuations deviating from the ideal pressure state could lead to production interruptions, equipment failures, or defective products, resulting in significant economic losses. Therefore, achieving real-time, precise pressure regulation has become key to improving the overall performance of the system. However, inherent limitations in system design, such as piping length, gas flow characteristics, and vacuum pump response, often result in significant time delays in pressure control. These delays not only affect system response time but can also compromise control performance and even lead to system instability. To overcome the slow response issue of traditional control methods when dealing with systems with large delays, Smith¹⁴ proposed a solution to compensate for dead-time. This approach, which is well-suited for open-loop stable processes, subsequently came to be known as the Smith predictor (SP). The SP compensates for time delays by predicting the output of a delay-free model. However, the classical SP is sensitive to perturbations, and low-pressure vacuum systems are subject to multiple perturbations, which limits its direct implementation in industrial applications.¹⁵ In response to the perturbation sensitivity of SP, extensive and in-depth research has been carried out by scientists both nationally and internationally. Some researchers have proposed the addition of disturbance observers, sliding mode control, and disturbance rejection filter units to the SP control strategy to compensate for or mitigate disturbances, improving the system's robustness and stability. Li and Gao¹⁶ proposed an improved Smith predictor-based decoupling control strategy with a disturbance observer to address the strong coupling, large time delays, multiple disturbances, and numerous uncertainties in compression refrigeration systems. Espín et al.¹⁷ presented an improved hybrid robust Smith predictor controller design, combining sliding mode methods, Smith predictors, and PD compensators, for processes with long dead times. Kumar and Ajmeri¹⁸ introduced an enhanced Smith predictor-based sliding mode control method for unstable processes with time delays, showing strong robustness against process parameter disturbances and load

disturbances. Chung et al.¹⁹ proposed a novel two-degree-of-freedom Smith predictor structure for improving PMSM speed control, with a sliding mode controller designed as the reference controller, enhancing disturbance rejection capabilities.

Disturbance observers and sliding mode control exhibit good theoretical performance, but certain limitations exist in their practical industrial applications. Disturbance observers are highly dependent on the model and require precise parameter tuning to ensure their performance, and these parameters are often difficult to obtain or adjust. The design of the sliding surface in sliding mode control is not always straightforward, especially for complex or nonlinear systems. These factors limit the direct application of these methods in industrial settings, with challenges arising in scenarios that demand high control accuracy and system stability, such as low-pressure control. Feliu-Batlle and Rivas-Perez²⁰ designed a PI-improved SP control structure for delayed systems in refinery control, which significantly improved the system robustness under varying parameters and disturbance conditions. In the research on disturbance suppression, Pereira et al.²¹ proposed a simplified filtered Smith predictor and extended feedforward control to address measurable disturbances in neonatal intensive care unit temperature control. This approach demonstrated significant advantages in disturbance suppression compared to other controllers. In the research on time-delay control of air-bounce systems, Li et al.²² introduced a second-order filter to enhance the SP for the time-lagged air-bounce control problem. This approach was combined with the BRGWO optimization algorithm to ensure optimal control performance under complex conditions, including uncertain delays, variable wind speeds, changes in stiffness, and drive disturbances. Zhang et al.²³ proposed an improved disturbance rejection controller utilizing the Smith predictor structure to compensate for errors caused by the speed filter in servo systems, aiming to enhance disturbance rejection capability. The structure is easy to implement and does not require parameter tuning, effectively mitigating the oscillatory speed response problem caused by the speed filter. Ali et al.²⁴ introduced a low-pass filter based on the Smith predictor to alleviate the impact of communication delays and synchronize inputs to the extended state observer. Experimental results demonstrated that the designed controller can effectively mitigate delays and disturbances.

Although various SP disturbance suppression schemes have been proposed for different industrial processes, pressure fluctuations in low-pressure systems are typically small, and even slight variations in control signals can significantly affect the stability and performance of the system. Existing control methods are difficult to directly apply to low-pressure control and still require improvements and optimizations for low-pressure vacuum systems that demand high real-

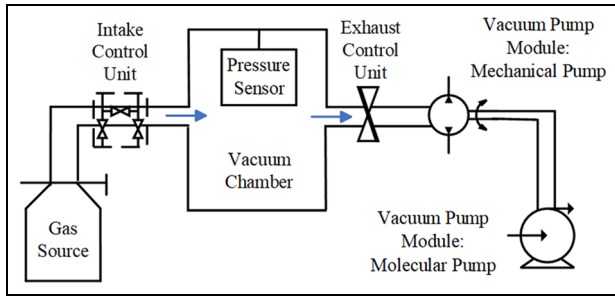


Figure 1. Schematic diagram of the low-pressure vacuum system.

time response. Therefore, this paper proposes an innovative solution for pressure control in low-pressure vacuum systems, such as vacuum coating, precision casting, high-performance composite material manufacturing, aerospace material preparation, and new energy material research and development. A composite controller (AFFPI) with adaptive tuning capabilities is designed, incorporating SP control and an anti-disturbance filter unit. By combining the anti-disturbance filter unit with SP control, the lag response and instability of the control system under external disturbances can be effectively reduced, enhancing the system's resistance to disturbances. At the same time, the composite controller, equipped with adaptive tuning functionality, can dynamically adjust control parameters based on varying process states, achieving faster and more accurate control.

The remaining chapter is organized as follows: The next section will provide a detailed description of the structure and components of the low-pressure vacuum system, followed by the establishment of the corresponding model. Section 3 will introduce the Smith predictor compensation control method for disturbance rejection in the pressure control of the low-pressure vacuum system. Section 4 will present simulation results to demonstrate the improvements in pressure regulation time and control signal stability when applying the proposed control method to handle various disturbances, followed by experimental validation and a brief analysis and discussion of the results. Finally, section 5 will summarize the contributions of this research and propose directions for future studies.

Low-pressure vacuum control system

As illustrated in Figure 1, the low-pressure vacuum system under consideration incorporates a series of components, including a gas source, an inlet control unit, a vacuum chamber, a pressure sensing device, an outlet control unit (i.e. a butterfly valve), a pump module, and a network of pipes connecting these modules. The gas source and the inlet control unit collaborate to regulate the flow of gas into the vacuum chamber, ensuring stability and controllability. The pumping module, a critical component of the system, employs a two-stage

pumping system comprising a mechanical pump and a molecular pump. These pumps collaborate to facilitate efficient vacuum extraction. The mechanical pump, functioning as the primary pump, initiates the process from atmospheric pressure or a lower vacuum level. The majority of the gas in the chamber is rapidly removed, establishing the pre-vacuum conditions. Following the initial evacuation by the mechanical pump, the molecular pump, functioning as the secondary pump, begins operation, further reducing the gas pressure to the ultra-high vacuum range. The pressure sensing device continuously monitors the internal pressure of the chamber. The exhaust control unit, as the core component of the low-pressure vacuum system, dynamically adjusts the closure of the butterfly valve based on signals from the controller and pressure sensing device. The gas extraction volume of the pumping module plays a crucial role in achieving precise control of the chamber pressure. The closed-loop control mechanism ensures stable and efficient system operation, with high resistance to interference.

For the low-pressure vacuum system shown in Figure 1, based on experimental data and the research conducted in the literature,²⁵ the system model can be approximated as a first order plus dead time (FOPDT) form, as shown in equation (1):

$$G(s) = \frac{K_c}{Ts + 1} e^{-Ls} = G_r(s) e^{-Ls} \quad (1)$$

Where T denotes the system's time constant, L signifies the dead time, and K_c represents the system's gain coefficient. This model effectively captures the system's fundamental dynamic characteristics. However, in contrast to conventional static first-order lag elements, the gain coefficient K_c of this system is not constant, but rather undergoes dynamic adjustments in response to variations in the closing of the exhaust control unit (e.g. valve opening). This characteristic renders the low-pressure vacuum system more vulnerable to the impact of operating conditions, load variations, and other external factors, thereby complicating the control process and increasing its susceptibility to instability.

Anti-disturbance Smith predictive compensation control method

AFFPI controller

To achieve rapid and stable pressure regulation in the low-pressure vacuum system, this paper proposes a controller with parameter adaptive adjustment functionality (referred to as AFFPI), which integrates the advantages of both feedback and feedforward control. The feedback control automatically corrects errors, ensuring steady-state performance and high adaptability. The feedforward control provides a fast response, minimizing steady-state errors, and preventing

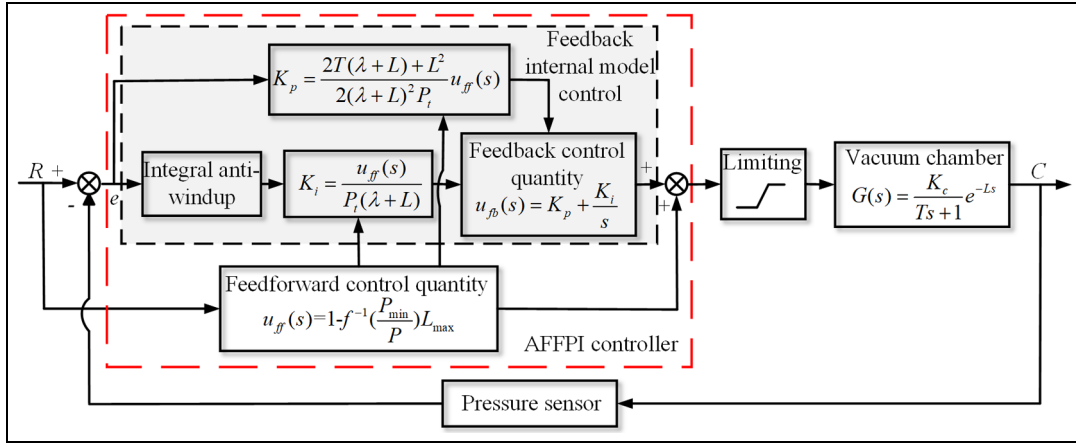


Figure 2. AFFPI controller.

feedback delays. With this design, the system can dynamically optimize parameters and achieve fast response coupling synchronization. The configuration of the control system is shown in Figure 2.

The total control value of the system is the sum of the feedforward predictive control value and the feedback internal model control value. The target pressure corresponding to the closing of the exhaust control unit valve under different process conditions is predicted using the derived formula (2), which is used as the feedforward control quantity for the controller. This is based on the characteristics of the control valve and gas flow²⁶:

$$u_{ff}(s) = 1 - f^{-1}\left(\frac{P_{\min}}{P}\right) L_{\max} \quad (2)$$

In the equation, $u_{ff}(s)$ is the feedforward control quantity of the exhaust control unit corresponding to the target pressure P at a fixed intake flow rate, $P_{\min}$ is the minimum pressure in the vacuum chamber when the exhaust control unit is fully open, and $L_{\max}$ refers to the maximum opening of the exhaust control unit, $f(\dot{s})$ is a nonlinear mapping from the target pressure to the valve closure.

The feedback control utilizes a PI controller based on internal model control. In consideration of the characteristics of the controlled object, this paper proposes an enhanced tuning method for the PI controller parameters. This enhanced method is derived from the Mac PID parameter tuning method²⁷ and is subsequently applied to the nonlinear low-pressure vacuum system illustrated in Figure 1:

$$K_p = \frac{2T(\lambda + L) + L^2}{2(\lambda + L)^2 P} u_{ff}(s), \quad K_i = \frac{u_{ff}(s)}{P(\lambda + L)} \quad (3)$$

In the equation, T represents the time constant of the first-order system, L denotes the delay time, and λ

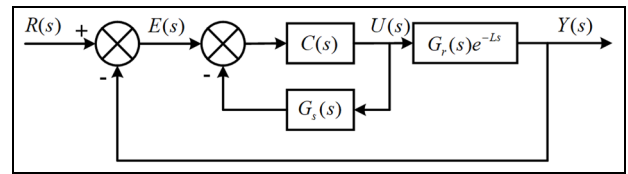


Figure 3. Pressure control structure of the low-pressure vacuum system with Smith predictor control.

is the filtering coefficient. The feedback control quantity of the controller is given by:

$$u_{fb}(s) = \frac{2T(\lambda + L) + L^2}{2(\lambda + L)^2 P} u_{ff}(s) + \frac{u_{ff}(s)}{Ps(\lambda + L)} \quad (4)$$

The enhanced proportional and integral gains, herein referred to as K_p and K_i , respectively, undergo dynamic recalibration, thereby facilitating autonomous tuning in accordance with disparate target control pressures and their associated feedforward predictive control quantities. This dynamic tuning mechanism enables flexible adaptation to changing operating conditions. In contrast to the Mac PID parameter tuning method, which relies on a static set of K_p and K_i values, the proposed method offers enhanced adaptability in dynamic environments.

Anti-disturbance Smith predictor compensation control

The conventional Smith predictor compensator is principally designed to compensate for delay time by incorporating a compensation loop in parallel with the controller. The pressure control structure of the low-pressure vacuum system with Smith predictor control is illustrated in Figure 3. In this figure, $G_r(s)e^{-Ls}$ represents the controlled object, that is, the low-pressure vacuum system, the main controller $C(s)$ is a PID

controller, and $G_s(s)$ represents the Smith predictor compensator.

To mitigate the effects of the time-delay term e^{-Ls} , the control structure includes the $G_m(s)$ loop. The characteristic equation of the closed-loop system can be expressed as:

$$1 + C(s)(G_s(s) + G_r(s)e^{-Ls}) = 0 \quad (5)$$

By defining $G_s(s) = G_r(s)(1 - e^{-Ls})$, equation (5) can be rewritten as:

$$1 + C(s)G_r(s) = 0 \quad (6)$$

As demonstrated in equation (6), following the implementation of delay compensation, the term e^{-Ls} is effectively excluded from the closed-loop characteristic equation, thereby exerting no influence on the system's stability. This process leads to the elimination of the delay effect, thereby addressing the issue of slow response, that is, caused by the large delay characteristics of the low-pressure vacuum system. Consequently, this enhances the efficiency of pressure regulation.

In this study, to achieve fast and stable pressure regulation of the low-pressure vacuum system, the traditional PID controller in the Smith predictor control is replaced with the AFFPI controller. The goal is to improve the pressure control performance of the low-pressure vacuum system. Moreover, considering that the actual system may be affected by external disturbances, internal model uncertainties, and uncertain variations in delay, and recognizing the sensitivity of the Smith predictor to disturbances, model uncertainties are treated as part of the external disturbance. When the model of the system changes, the resulting output change due to the model variation is approximated as an external disturbance caused by a hypothetical feedback loop.

The present study proposes the incorporation of an anti-disturbance filtering unit within the Smith predictor, with the objective of engineering an anti-disturbance Smith predictor controller. The efficacy of this approach is predicated on the judicious calibration of a designated filtering gain factor, a maneuver that has the potential to enhance the disturbance rejection capability of the predictor when confronted with a myriad of disturbances. The control structure is illustrated in Figure 4. In this figure, $F(s)$ represents the anti-disturbance filtering unit, $D(s)$ denotes the external disturbance, $G_m(s)$ is the behavior prediction model of the low-pressure vacuum system without considering delay factors, L and L_0 , respectively represent the actual and parameter identification-derived time delays of the low-pressure vacuum system.

Considering the impact of external disturbances and uncertain variations in delay on the low-pressure vacuum system, the controlled object can be represented as:

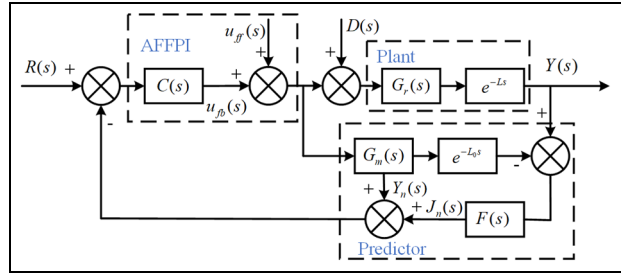


Figure 4. Anti-disturbance Smith predictor control system structure.

$$\begin{cases} P(s) = G_r(s)e^{-Ls} \\ G_r = G_m + \Delta G, L = L_0 + \Delta L \end{cases} \quad (7)$$

In the equation: ΔG represents the error between the actual system and the predicted model under external disturbances; ΔL represents the variation in delay.

The closed-loop transfer function of the anti-disturbance Smith predictor control system is:

$$G(s) = \frac{Y(s)}{R(s)} = \frac{C(s)G_r(s)e^{-Ls}}{1 + C(s)G_m(s) + C(s)F(s)(P(s) - P_m(s))} \quad (8)$$

In the equation: $P_m(s) = G_m(s)e^{-L_0s}$ represents the behavior prediction model of the low-pressure vacuum system, considering the delay factors.

The anti-disturbance filtering unit is:

$$F(s) = \frac{1}{\tau s + 1}, \tau > 0 \quad (9)$$

In the equation: τ is the adjustable filter gain factor.

1. In the absence of external disturbances and if the delay uncertainty remains constant, there exists $P(s) = P_m(s)$. According to equation (8), the characteristic equation of the closed-loop system can be simplified as follows:

$$1 + C(s)G_m(s) = 0 \quad (10)$$

2. In the event of delay uncertainty, that is, when $\Delta L \neq 0$ occurs, as indicated by equation (8), the characteristic equation of the closed-loop system can be expressed as follows:

$$1 + C(s)G_m(s) + C(s)F(s)G_r(s)(e^{-Ls} - e^{-L_0s}) = 0 \quad (11)$$

3. In the event of an external disturbance, that is, $\Delta G \neq 0$, the transfer function representing the

influence of the external disturbance on the system output can be expressed as:

$$\frac{Y(s)}{D(s)} = \frac{P(s)(1 + C(s)G_m(s) - C(s)F(s)P_m(s))}{1 + C(s)G_m(s) - C(s)F(s)P_m(s) + C(s)F(s)P(s)} \quad (12)$$

The closed-loop system characteristic equation under external disturbance can be obtained from equation (12) as:

$$1 + C(s)G_m(s) - C(s)F(s)(G_m(s) - G_r(s))e^{-Ls} = 0 \quad (13)$$

A theoretical analysis of equations (11) and (13) reveals that the designed disturbance-resistant Smith predictor optimizes the system error caused by external disturbances and delay uncertainties through its built-in disturbance-resilient filtering unit $F(s)$. This filtering unit efficiently eliminates high-frequency disturbance signals, thereby optimizing and reconstructing of the feedback signal. This method has been demonstrated to substantially reduce the impact of delay uncertainties and external disturbances on the system's pressure control accuracy, thereby significantly enhancing the control performance of the low-pressure vacuum system.

Control system stability analysis

To verify the stability of the proposed control system, the closed-loop stability of the designed disturbance-resistant Smith predictor control system is analyzed.²² It is assumed that the low-pressure vacuum system $G_r(s)$ without delay is a stable system, and that the main controller $C(s)$ and filter $F(s)$ also remain stable. The sufficient and necessary condition for the stability of the closed-loop system in Figure 4 is as follows:

$$|C(s)|H + |C(s)||F(s)||\Delta P(s)| < 1 \quad (14)$$

In the equation, $\Delta P(s)$ represents the lumped disturbance of the system (including external disturbances and uncertain delays) and H is the upper bound of the lumped disturbance:

$$|\Delta P(s)| = |G_r(s)e^{-Ls} - G_m(s)e^{-L_0s}| < 1 \quad (15)$$

There also exists:

$$|C(s)| < \|C(s)\|_\infty, |F(s)| < \|F(s)\|_\infty \quad (16)$$

Substituting equations (15) and (16) into equation (14) and simplifying, we obtain:

$$|\Delta P(s)| < \frac{1}{1 + \|F(s)\|_\infty} \frac{1}{\|C(s)\|_\infty} \quad (17)$$

According to equation (17), it can be determined that when the upper bound of the lumped disturbance for the low-pressure vacuum system is designated as $1/(\|C(s)\|_\infty(1 + \|F(s)\|_\infty))$, the closed-loop control system of the designed disturbance-resistant Smith predictor compensation controller is stable.

Theorem 1: In the absence of uncertain delay variation ($\Delta G \neq 0$, $\Delta L = 0$), external disturbances can be represented as a lumped disturbance of the system. This postulation is supported by the following theorem:

$$\Delta P(s) = (G_r(s) - G_m(s))e^{-Ls} \quad (18)$$

Substituting equation (17) into this postulation, the stability condition of the closed-loop system under external disturbance is obtained as:

$$|\Delta G(s)| < \frac{\|e^{-Ls}\|}{1 + \|F(s)\|_\infty} \frac{1}{\|C(s)\|_\infty} \quad (19)$$

Theorem 2: In the absence of external disturbances, but in the presence of uncertain delay variations ($\Delta G = 0$, $\Delta L \neq 0$), the lumped disturbance of the system can be expressed as follows:

$$\Delta P(s) = G_r(s)(e^{-Ls} - e^{-L_0s}) \approx G_r(s)(L_0 - L)s = G_r(s)\Delta Ls \quad (20)$$

Substituting equation (17) into this equation yields the stability condition of the closed-loop system under uncertain delays:

$$|\Delta L(s)| < \frac{1}{1 + \|F(s)\|_\infty} \frac{1}{\|C(s)\|_\infty \|G_r(s)\|_\infty} \quad (21)$$

Simulation and experimental research

To verify the performance of the proposed control method, a low-pressure vacuum system was designed and constructed, as shown in Figure 5, based on the schematic diagram in Figure 1. The parameters of key experimental equipment are listed in Table 1.

The vacuum chamber of the system has a volume of ~40 L, and the exhaust control unit utilizes a butterfly valve. During pressure control, the system regulates the inlet flow rate through the mass flow meter, while the amount of gas extracted from the chamber by the pump is controlled by adjusting the valve plate position, thus ensuring precise pressure control. The valve plate position is dynamically computed using the anti-

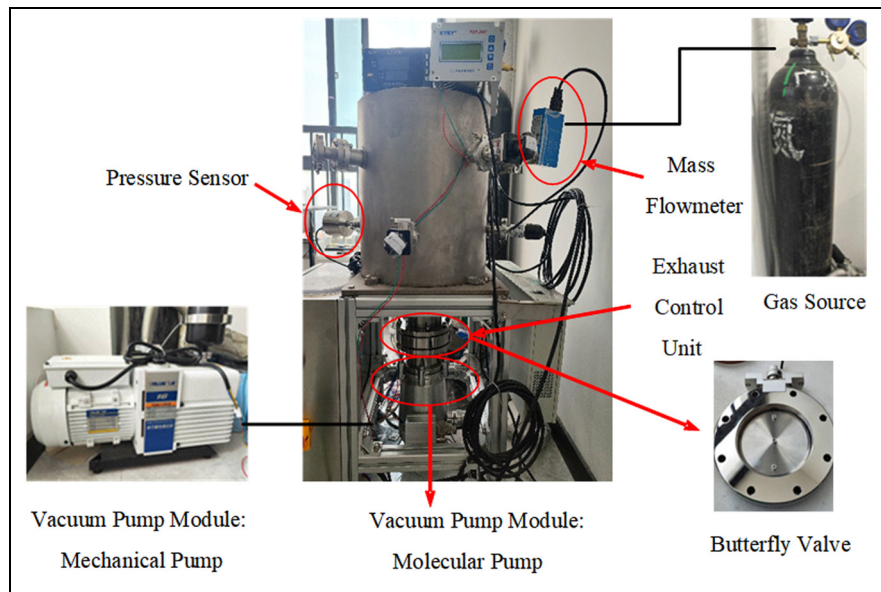


Figure 5. The low-pressure vacuum control system.

Table 1. Parameters of some experimental equipment.

Equipment	Parameter	
Mass flowmeter (model: LF420-S)	Control range	0–500 sccm
	Control accuracy	±1% F.S.
	Linearity	±0.5 F.S.
	Repeatability	±0.2 F.S.
Butterfly valve (model: GC-DFI00-KBR)	Pressure control accuracy	±0.5 Pa
	Valve body leak rate	$\leq 1.0 \times 10^{-12} \text{ Pa m}^3/\text{s}$
Mechanical pump (model: VRD-16)	Pumping speed	16 m ³ /h
	Ultimate pressure	$4.0 \times 10^{-1} \text{ Pa}$
Molecular pump (model: FF-100/110)	Rated speed	1440 rpm
	Pumping speed	110 L/s
	Ultimate pressure	$\leq 6.0 \times 10^{-8} \text{ Pa}$
Pressure sensor (model: CMR 364)	Rated speed	42,300 rpm
	Measuring range	0.01–110 Pa
	Precision	0.2% of measurement
	Resolution	0.003% F.S.

perturbation Smith predictive control algorithm proposed in this paper. This algorithm is implemented on a control unit based on the STM32 microcontroller, which guarantees real-time, precise adjustment of the valve plate angle. Additionally, a stepper motor is used to drive the rotation of the valve plate, thereby improving both the precision and response speed of the control system.

Simulation of anti-disturbance Smith predictor compensation control performance

To verify the control performance of the proposed Anti-disturbance Smith predictor controller, a step response method was employed under the condition of an intake flow rate of 150 sccm. The curve of pressure variation with respect to time was obtained when a step

change was applied to the valve under these process conditions. Subsequently, system parameters were identified using the least squares method, leading to the FOPDT model of the low-pressure vacuum system, as described in equation (22). Control system performance simulations were then conducted in the MATLAB/Simulink environment:

$$G(s) = \frac{70.7577}{9.135s + 1} e^{-1.21s} \quad (22)$$

Under the same target pressure and control accuracy requirements, the efficacy of various control methodologies was evaluated under conditions of no external disturbances and uncertain time delays. The control methods compared include the feedforward–feedback composite control (AFFPI) proposed in this paper, the

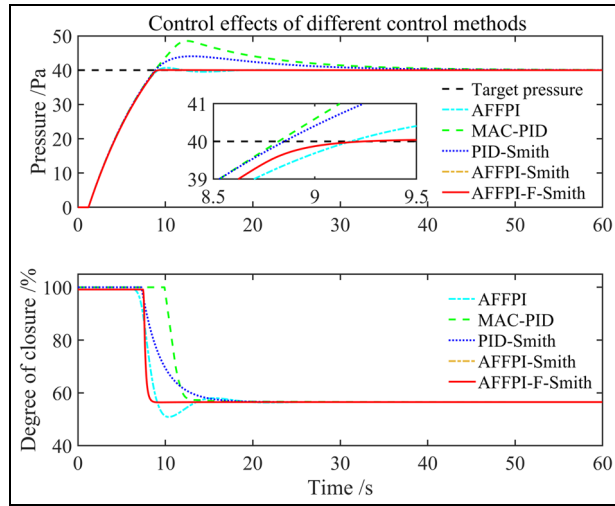


Figure 6. Pressure control performance under the conditions of no external disturbances and uncertain time delays.

Table 2. Comparison of pressure control performance results for different control methods.

Control method	Adjustment time, t/s (control accuracy 0.1 Pa)	Overshoot (%)	ITAE
AFFPI	17.76	1.73	551.80
Mac-PID	55.20	20.31	2554.61
PID-Smith	49.41	10.13	1658.63
AFFPI-Smith	9.06	0.13	508.74
AFFPI-F-Smith	9.06	0.13	508.74

classical PID control (Mac-PID), the PID control with Smith predictor (PID-Smith), the AFFPI control with Smith predictor (AFFPI-Smith), and the Anti-disturbance Smith predictor compensation control method (AFFPI-F-Smith). The pressure and valve closure response curves under different control methods are shown in Figure 6.

The performance evaluation metrics for each control method were extracted based on the pressure response curve, and the results are presented in Table 2. The evaluation metrics include settling time, overshoot, and ITAE (Integral of Time-weighted Absolute Error) under the same control accuracy (0.1 Pa). The results indicate that the classical PID control and the PID control with Smith predictor exhibit relatively long settling times, larger overshoots, and higher ITAE values. In contrast, the AFFPI control method proposed in this paper, after the introduction of the Smith predictor controller, exhibited a substantial enhancement in pressure control performance, as evidenced by an average reduction in settling time by 31.73 s, a decrease in overshoot by 10.59%, and an average reduction in ITAE by 1619.41.

To further validate the anti-disturbance performance of the control method proposed in this paper, four different simulation test cases were conducted. These cases

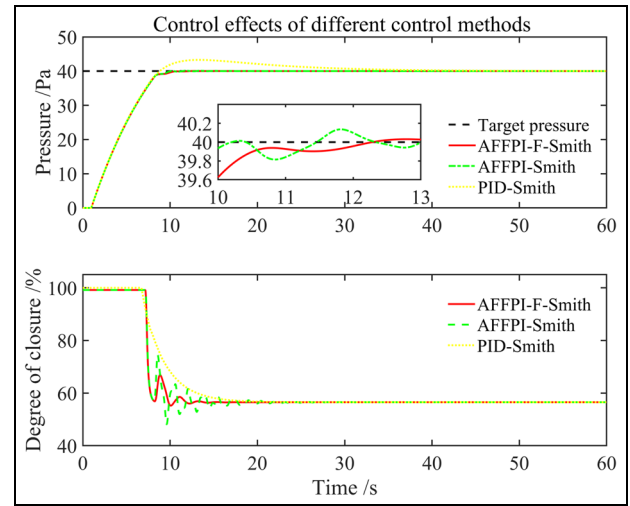


Figure 7. Pressure control performance when the time delay variation is 20%.

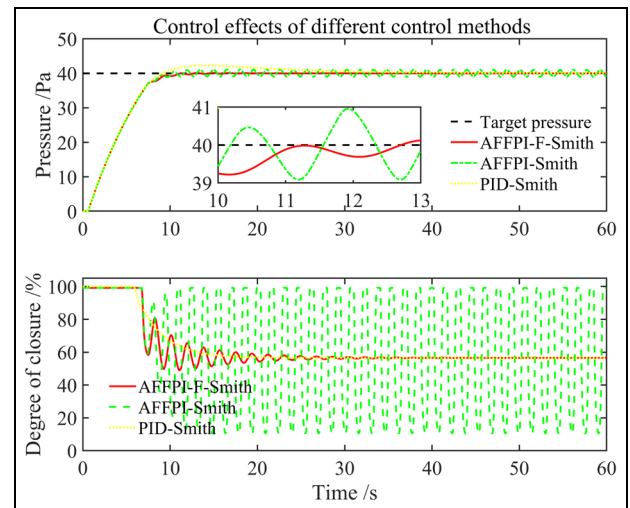


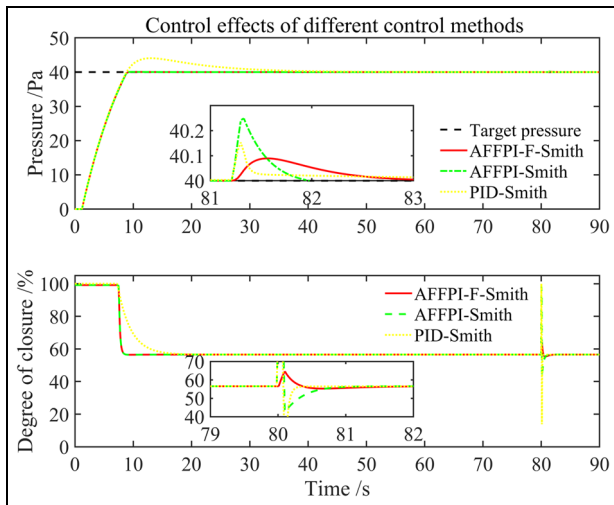
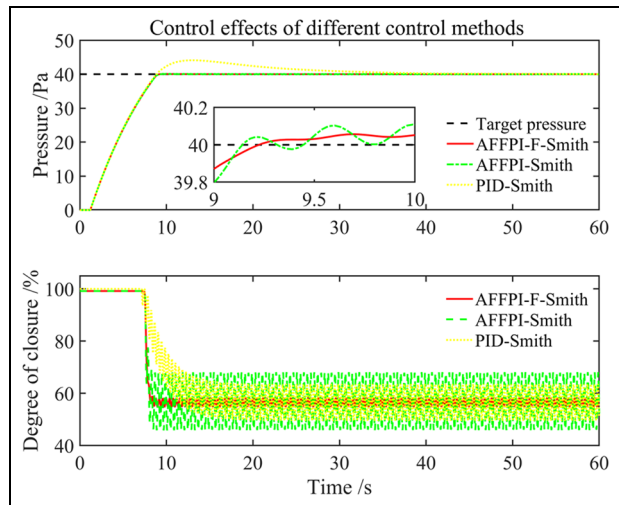
Figure 8. Pressure control performance when the time delay variation is 50%.

included scenarios with time delay variations of 20% and 50%, as well as the introduction of step and sinusoidal disturbance signals into the control loop. The time delay uncertainties were used to simulate situations where the response delays of sensors, valves, and pipelines do not align with the parameter identification results. The step signal was employed to simulate instantaneous disturbances, such as abrupt changes in gas flow due to valve switching operations or gas leaks. The sinusoidal signal was used to simulate periodic disturbances, such as recurring pressure fluctuations caused by equipment, pump vibrations, or temperature variations.

In the simulation tests, the dynamic response characteristics of pressure and valve closure were compared and analyzed under different control methods (as shown in Figures 7–10). Key performance evaluation metrics were extracted from the response curves to

Table 3. Comparison of pressure control performance results for different control methods under four test cases.

Perturbation	Control method	Adjustment time, t/s (control accuracy 0.1 Pa)	CV	TV	IAE	ITAE
20% delay variation	PID-Smith	48.18	0.0134	0.44	219.11	1444.20
	AFFPI-Smith	12.00	0.0125	1.71	172.92	477.56
	AFFPI-F-Smith	10.53	0.0124	0.72	173.09	478.77
50% delay variation	PID-Smith	46.20	0.0123	0.43	192.70	1153.23
	AFFPI-Smith	-(Oscillation)	0.1353	106.85	189.31	1478.46
	AFFPI-F-Smith	16.92	0.0121	2.67	161.35	451.43
Step signal	PID-Smith	81.36	0.0143	2.16	238.21	1683.08
	AFFPI-Smith	81.54	0.0130	1.54	181.52	518.42
	AFFPI-F-Smith	9.06	0.0128	0.62	181.48	514.96
Sine signal	PID-Smith	52.77	0.0164	64.67	237.88	1659.67
	AFFPI-Smith	10.43	0.0185	102.03	182.99	566.68
	AFFPI-F-Smith	9.06	0.0129	16.43	181.62	517.33

**Figure 9.** Pressure control performance under step load disturbance.**Figure 10.** Pressure control performance under sine wave load disturbance.

characterize control performance. These metrics include the settling time required to achieve a control accuracy of 0.1 Pa, integral of absolute error (IAE), integral of time-weighted absolute error (ITAE),²⁸ total variation (TV) of the control signal, and control variance (CV).²¹ The quantized results of these performance metrics, as evaluation standards, are summarized in Table 3, providing data support for the comprehensive performance evaluation of the control methods.

As demonstrated in Figures 7 and 8, the classical PID-Smith control method demonstrates a certain degree of robustness in dealing with time delay uncertainties, maintaining a relatively stable valve closure and pressure fluctuation under varying levels of time delay uncertainty, and achieving a lower TV metric. However, it still suffers from issues such as longer settling time, and higher IAE and ITAE values. This hinders the method's ability to meet the efficiency and accuracy requirements of pressure control in low-pressure vacuum systems. The AFFPI-Smith method demonstrates improvements in settling time, CV, IAE, and ITAE within a certain range of time delay

variations when compared to the PID-Smith method. However, in scenarios with large time delay variations, oscillations may occur, leading to increased CV and TV metrics, thereby failing to achieve the target control accuracy.

In contrast, the AFFPI-F-Smith method proposed in this paper demonstrates shorter pressure control times, superior CV, IAE, and ITAE metrics, and better adaptation to time delay uncertainties caused by external disturbances, compared to both PID-Smith and AFFPI-Smith methods under various levels of time delay variations.

As demonstrated in Figures 9 and 10, the experimental results indicate that both the PID-Smith and AFFPI-Smith control methods manifest substantial pressure disturbance deviations after the introduction of a step disturbance (at 80 s) and a sinusoidal disturbance. This phenomenon culminates in the disruption of the system's steady-state and a decline in control performance. In contrast, the AFFPI-F-Smith method proposed in this paper demonstrates enhancements in performance metrics such as settling time, CV, TV,

IAE, and ITAE. This method not only fulfills the efficiency and accuracy requirements for pressure control in low-pressure vacuum systems but also enhances the anti-disturbance capability of the control system.

According to the data presented in Table 3, the AFFPI-F-Smith method exhibited the optimal overall control performance across the four test cases when compared to other control methods. This method demonstrated the shortest control time and superior CV, IAE, and ITAE metrics. In comparison to the AFFPI-Smith method, the AFFPI-F-Smith method improves the anti-disturbance performance by an average of 19.66%.

Anti-disturbance Smith prediction compensation control performance experiment

During the construction of the low-pressure vacuum control system experimental platform, preliminary tests revealed several unavoidable disturbance factors, such as response delay errors in devices like sensors, valves, and pipelines, gas leakage anomalies, pump vibrations, and temperature fluctuations. These factors led to significant dynamic fluctuations in the pressure control process, causing the gas outlet control unit valve to frequently exceed the designed dead-zone range. To comprehensively evaluate the pressure control performance and disturbance mitigation capabilities of the proposed control method, systematic pressure control experiments were conducted on the constructed experimental platform, using simulation results from previous research.

The experimental setup involved initializing the inlet flow rate of the low-pressure vacuum system to 100 sccm. Under the same target pressure and control accuracy requirements, the control effects of the basic feedforward and feedback control method (AFFPI), classical PID control method (Mac-PID), PID control method with Smith prediction (PID-Smith), AFFPI control method with Smith prediction (AFFPI-Smith), and anti-disturbance Smith prediction compensation control method (AFFPI-F-Smith) were compared. The pressure and valve closure response curves of the five control methods in the actual system are shown in Figure 11. Relevant performance evaluation metrics were obtained from the response curves, including the settling time, CV, TV, IAE, and ITAE corresponding

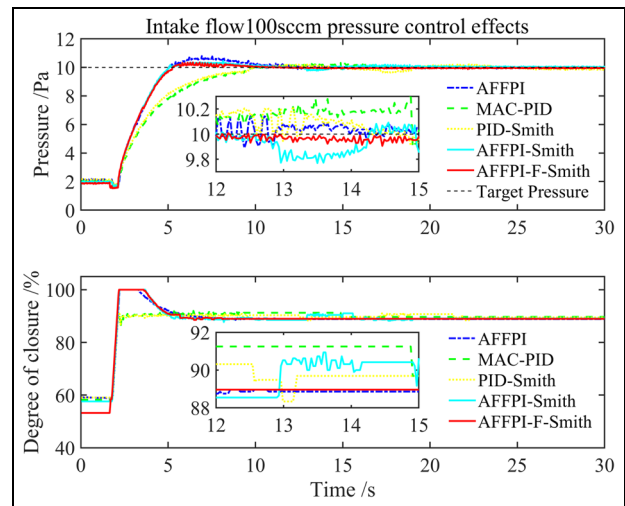


Figure 11. Pressure control performance of five control methods when the inlet flow rate is 100 sccm.

to each method under the same control accuracy (0.2 Pa). The CV and TV indices were calculated after the influence of the feedforward control had ended. The parameter values of the relevant indicators are provided in Table 4.

As demonstrated in Figure 11 and the data presented in Table 4, under conditions wherein the inlet flow rate of the low-pressure vacuum system is set at 100 sccm and the accuracy requirement is set at 0.2 Pa, the pressure control method proposed in this paper exhibits an average reduction of 7.32 s in settling time when compared to the four control methods: AFFPI, Mac-PID, PID-Smith, and AFFPI-Smith. Furthermore, the CV, TV, ITE, and ITAE indices demonstrate average reductions of 1.8247×10^{-5} , 0.385, 3.85, and 26.23, respectively, thereby substantiating substantial enhancements in control accuracy, settling time, and stability. Specifically, when compared to the Smith prediction control method (AFFPI-Smith) that does not account for disturbances, the ITAE is reduced by 4.72, and the anti-disturbance performance is enhanced by 11.12%.

To provide further validation of the pressure control performance of the proposed control method under scenarios involving fluctuations in process parameters and changes in target pressure, experiments were conducted with four sets of operating conditions. In these experiments, the inlet flow rates were set to 50, 70, 120, and

Table 4. Comparison of pressure control performance results for five control methods when the inlet flow rate is 100 sccm.

Control method	Adjustment time, t/s (control accuracy 0.2 Pa)	CV/ 10^{-5}	TV	IAE	ITAE
AFFPI	11.58	5.2575	0.22	6.83	43.24
Mac-PID	16.17	4.9490	0.51	12.31	74.19
PID-Smith	22.23	2.7333	0.58	12.38	95.95
AFFPI-Smith	13.62	4.3868	0.55	5.96	42.45
AFFPI-F-Smith	8.58	2.5070	0.08	5.52	37.73

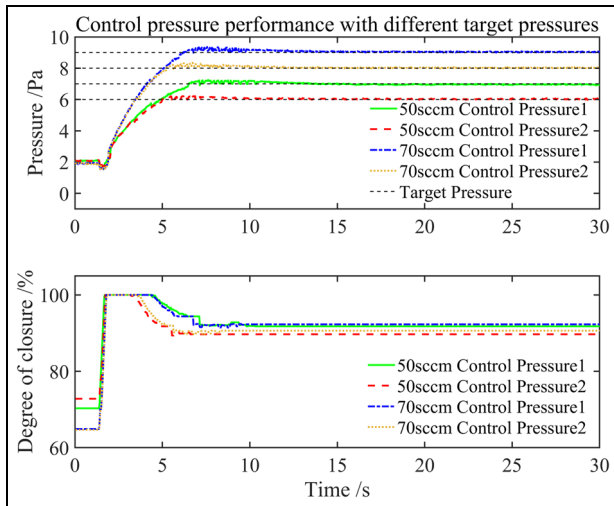


Figure 12. Pressure control performance with different target pressures when the inlet flow rate is 50 and 70 sccm.

150 sccm, respectively. For each set of conditions, two different target pressure values were established. The control performance of the proposed method was then systematically examined under these dynamic conditions. The ensuing experimental results are presented in Figures 12 and 13.

The experimental results demonstrate the effectiveness of the proposed pressure control method. The method maintains consistent control performance, even when faced with significant changes in system operating conditions or dynamic adjustments to the target pressure. Specifically, once steady state is achieved, both the output pressure and valve closure degree exhibit minimal fluctuations, confirming the method's robustness and anti-disturbance performance under complex operating conditions. These findings highlight the theoretical and practical significance of the proposed method, offering valuable insights for the design and optimization of industrial control systems.

Conclusion

To address the significant time-delay characteristics and multiple disturbance issues in the low-pressure vacuum system, and to achieve efficient and high-precision pressure control, an anti-disturbance Smith predictor compensation control method is proposed in this paper. Based on the dynamic characteristics of the low-pressure vacuum system and leveraging the synergistic advantages of feedforward and feedback control, a composite controller with parameter adaptive tuning functionality (AFFPI) was designed. This controller can optimize control parameters according to changes in the process state, ensuring rapid response and

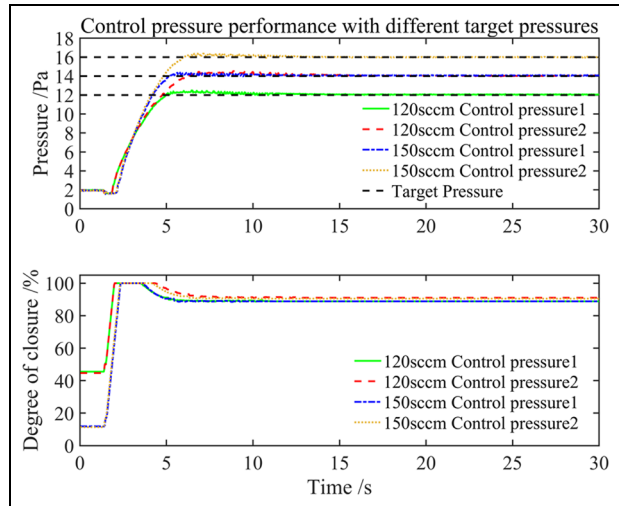


Figure 13. Pressure control performance with different target pressures when the inlet flow rate is 120 and 150 sccm.

maintaining good steady-state accuracy under various operating conditions. Additionally, to address the challenges posed by the inherent lag characteristics of the vacuum system on control performance, an internal control loop composed of a SP was designed based on the AFFPI, effectively solving the problem of slow response in the control system caused by the significant lag characteristics of the low-pressure vacuum system. Finally, an anti-disturbance filtering unit was introduced into the pressure feedback channel. This unit helps mitigate the fluctuations in control signals caused by disturbances to the SP, preventing the performance degradation typically observed in traditional SP control methods when disturbances are present, thus achieving efficient and high-precision pressure control in the low-pressure vacuum system.

The simulation results demonstrate that, in comparison to the AFFPI, Mac-PID, and PID-Smith methods, the proposed method reduces the average settling time by 31.73 s, decreases the average overshoot by 10.59%, and lowers the ITAE index by 1619.41 in the pressure control process. Under four disturbance conditions, its anti-disturbance performance improves by an average of 19.66% compared to the Smith predictor control method without disturbance compensation (AFFPI-Smith). The experimental findings further substantiate the efficacy of this approach, exhibiting an average reduction of 7.32 s in settling time when compared to alternative methods. Additionally, a decline in the CV, TV, ITE, and ITAE indices by 1.8247×10^{-5} , 0.385, 3.85, and 26.23, respectively, was observed, signifying enhanced control accuracy, regulation time, and stability. In particular, the ITAE index is reduced by 4.72, and the anti-

disturbance performance is enhanced by 11.12% when compared to the AFFPI-Smith control method. Moreover, under dynamic variations in inlet flow and target pressure, the proposed method maintains excellent control performance, with minimal fluctuations in steady-state pressure output and valve closure, demonstrating strong robustness and anti-disturbance capability against external disturbances.

Although the current AFFPI-F-Smith controller has some adaptive tuning capabilities for control parameters in the low-pressure vacuum control system, its adaptability may still lack sufficient flexibility when faced with complex and varying operating conditions. Future research could explore the incorporation of machine learning techniques (such as deep learning, reinforcement learning, etc.) into the adaptive control algorithm, allowing the controller to continuously optimize control strategies through historical data and real-time feedback. This data-driven control approach is expected to maintain good performance under a wider range of operating conditions and disturbances.

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Ethical considerations

This article does not contain any studies with human or animal participants.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The authors would like to thank the National Natural Science Foundation of China (No. 52005003), the Major Science and Technology Project of Anhui Province (No. 202203a05020041), and the Science and Technology Project of Wuhu (No. 2023pt08).

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

All relevant data are within the paper and can also contact the corresponding author on reasonable request.

Author Contributions

Lulu Wu: Conceptualization, Software, Resources, Writing, Editing, Funding Acquisition, Project Administration Licheng Huang: Methodology, Validation, Result Analysis, Data Curation, Writing Original Draft, Visualization Yuan Wang: Software, Supervision, Project Administration Benchu Jiang: Resource, Supervision, Funding Acquisition Dazhu Li: Supervision, Investigation.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

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Appendix

The list of nomenclatures

Abbreviations

AFFPI	Adaptive feedforward and feedback PI controller
BRGWO	Bidirectional random Gray Wolf optimizer
CV	Control variance
FOPDT	First order plus dead time
ITE	Integral of absolute error
ITAE	Integral of time-weighted absolute error
PI	Proportional–integral
PID	Proportional–integral–derivative
PMSM	Permanent magnet synchronous motor
sccm	Standard cubic centimeters per minute
SP	Smith predictor
TV	Total variation

Symbols

H	The upper bound of the lumped disturbance
K	Gain coefficient
K_c	The gain coefficient of the low-pressure vacuum system
K_p	Proportional gain
K_i	Integral gain
L	Dead time
L_0	Parameter identification-derived time delays
$L_{\max}$	The maximum opening of the exhaust control unit,
P	Target pressure
$P_{\min}$	The minimum pressure in the vacuum chamber
T	Time constant
λ	The filtering coefficient
τ	The adjustable filter gain factor
ΔG	The error between the actual system and the predicted model under external disturbances
ΔL	The variation in delay